

FLIGHT SIMULATION AND PROPULSION LAB

IV Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE11	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100
LIST OF EXPERIMENTS								
PROPULSION LAB 1. Stripping of a piston engine, visual inspection and reasoning for common troubles and trouble shooting 2. Study of piston engine (Port Timing Diagram) 3. Study of piston engine (Valve Timing) 4. Performance of piston engine 2-stroke 5. Performance of piston engine 4-stroke 6. Engine Balancing.								
FLIGHT SIMULATION LAB 1. Demonstration of cockpit display systems and simulator commands 2. Calculation of CL and CD using the Speed-Power Method 3. Effect of Flaps on Take-off and landing Performance 4. Effect of Weight on Take-off and landing Performance 5. Estimate the Distance travelled by the aircraft using GPS								
Reference Books: 1. H. Cohen, G. F. C. Rogers and H. I. H. Saravanamuttoo, "Gas Turbine Theory", Longman, 2006. 2. M. L. Mathur and R. P. Sharma, "Gas Turbine, Jet and Rocket Propulsion", Standard Publishers & Distributors, Delhi, 2007.								
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO 1. Illustrate the port and valve timing diagrams 2. Estimate the performance of four stroke petrol/diesel engines 3. Demonstrate the instrumentation layouts of a civil transport airplane 4. Evaluate the effect of aircraft weight on take-off /Landing performance 5. Analyze the effect of flaps on take-off performance of an airplane								